L.5. Models

Models are purpose dependent

What is it?

Purpose dependence means that a model is only “good” or “bad” **for a specific goal**.

Since all models simplify reality, we cannot judge them in general.

A model is acceptable if its simplifications **do not harm its ability to do the job we need it to do**.

Example:

- **Bohr’s atom model:**

  Not true as a full theory of atoms, but still very helpful for teaching basic atomic structure.

  Good for learning, not good for making precise scientific predictions.

- **Subway map:**

  Shows stations but ignores real distances and street layout.

  Great for planning metro trips, bad for walking around the city.

- **Crash test dummy:**

  Not like a real human in many ways, but good for testing how people might be injured in car crashes.

How is a model different from a theory?

What is the difference?

A **theory** aims to give a *true and complete description* of how some part of the world works.

If a theory is shown to be false, we **reject it**.

A **model**, on the other hand, is allowed to be **false or simplified** as long as it is **useful for a certain purpose**.

We judge a model by how well it works for that purpose, not by whether it describes everything correctly.

In short:

- **Theory:** must be true or accurate in its claims.
- **Model:** may be idealized or simplified—what matters is usefulness.

Examples:

- **Bohr's atom model:**

  As a *theory*, it was rejected when proven false.

  As a *model*, it is still used today because it is useful for teaching.

- **Subway map:**

  Not a true picture of the city, but a very useful model for travel planning.

  A theory could not be this inaccurate.

Positive analogy

What is it?

A positive analogy is a **similarity** between the model and the target.

It is a feature they both share.

Why does it occur in models?

Positive analogies occur because a model must share some important properties with the target in order to represent it and give useful information. Without similarities, the model would tell us nothing about the real system.

Example:

A scale model airplane and a real airplane both have **similar wing shapes**.

A rat and a human both have **similar basic organs and hormones**.

Negative analogy

What is it?

A negative analogy is a **difference** between the model and the target.

It is a feature the model has that the target does not, or the other way around.

Why does it occur in models?

Negative analogies occur because all models **simplify or idealize** reality.

They must differ from the target in some ways, otherwise they would be too expensive, too complex, or impossible to use.

Example:

A scale model airplane is **much smaller** than the real airplane.

Rats are **short-lived** compared to humans.

Neutral analogy

What is it?

A neutral analogy is a property of the model where we **do not yet know** if the target has the same property.

It is an “unknown similarity.”z

Why does it occur in models?

Neutral analogies occur because we often use models to **explore or test new ideas**.

When we manipulate a model, new behaviours appear, and we must check if the real target system behaves the same way.

Example:

A drug works well in rats (model), but we **do not yet know** if it works the same in humans (target).

In Schelling's segregation model, groups cluster but we **don't know** if real cities behave that way without more evidence.

Models vs. Experiments

Similarities:

In both models and experiments, we **set variables and parameters**, we **manipulate** the system, and we **observe** what happens as a result. This means models mimic the basic structure of experimental control, intervention, and measurement.

Differences:

In models, **internal validity** is usually not a big problem because the modeler creates the system and knows exactly how the result is produced. In experiments, internal validity is a major concern because real-world factors can interfere, and we must be sure that the observed effect truly comes from the intervention.

Models also require us to **justify neutral analogies,** we must explain why what happens in the model should also happen in the real target. This is a challenge for all models, but only for some experiments (especially experiments done *not* directly on the real target).

What makes a good model?

Similarity

What is it?

Similarity means that the model shares several important properties with the target.

These shared properties are **positive analogies**, and they must be **relevant for the purpose** of the model.

Examples:

A scale model airplane has **similar wing shape and airflow behaviour** to a real airplane, which makes it useful for studying aerodynamics.

A crash test dummy has **similar body shape and weight distribution** to a human body, which makes it useful for studying crash impacts.

Robustness

What is it?

Robustness means that a model's result stays the same even if you change one of its assumptions.

If the result does not change, that assumption is **not important** for the model's purpose.

Example:

If a mathematical model gives the same outcome whether a parameter is 0.3 or 0.5, the result is **robust** to that parameter.

If changing the exact shape of an airplane wing in a simulation barely affects the predicted lift, the lift result is **robust** to that assumption

Parameter precision

What is it?

Parameter precision means that one model gives a **more detailed or specific** description of its parameters than another.

If the precise model's parameter description **automatically includes** the less precise one, then the first model has higher parameter precision.

Examples:

A model that says the rate of change is **a + b·N** is more precise than a model that only says it is **some function of N**, because a linear function is a specific kind of function.

A model that states a reaction happens at **exactly 25°C** is more precise than one that says it happens at **room temperature**.

Simplicity

What is it?

Simplicity means that a model uses **fewer parameters** or fewer details than another model.

A simpler model is easier to understand and work with, but may be less detailed.

Example:

A model that uses **one parameter** to describe growth is simpler than a model that uses **three parameters**.

A straight-line model is simpler than a curved model because it requires fewer assumptions.

Tractability

What is it?

Tractability means that the model can be **worked with and solved** using reliable methods.

- A model is **computationally tractable** if a computer can calculate its result efficiently.
- A model is **analytically tractable** if you can get the result directly through logical or mathematical deduction.
- A model is **theoretically tractable** if it is either computationally or analytically tractable *and* the steps can be justified using a background theory.

Example(s):

A simple regression model is analytically tractable because you can solve it using algebra.

A simulation of many interacting particles is computationally tractable if the computer can solve it in reasonable time.

A chemical model using Lewis structures is theoretically tractable because it can be solved using functional group theory.

Transparency

What is it?

Transparency means that the user can **understand how the model produces its result**.

If the model's steps are clear enough for a person to follow mentally, it is transparent. If the model is too complex to understand without blindly trusting a computer, it is not transparent.

Example(s):

A simple mathematical model where you can follow each calculation step is transparent.

A large brain simulation with millions of interacting units is not transparent, because no person can fully understand how the final output is produced.

Mirror models

What is it?

Mirror models are models that try to be **as similar as possible** to the real target.

They include many details so the model “mirrors” the real system closely.

Example:

A detailed epidemic simulation that includes real population data, household sizes, workplaces, and movements aims to mirror a real city.

A finite element model of a car body that closely matches the car’s exact shape and material properties is also a mirror model.

Limitations of “models as mirrors”

1. **They become very complex**

   Adding many details makes the model hard to understand, hard to work with, and less transparent.

2. **They lose simplicity and tractability**

   The more details you include, the harder it becomes to calculate results or apply theory.

   You often need heavy computation, and you can’t easily see why a result happens.

3. **They still contain idealizations**

   Even highly detailed models *still* require assumptions, shortcuts, and missing information.

   So they are never truly perfect mirrors.

4. **External validity remains a problem**

Even if the model is very similar to the target, you still need to justify that the assumptions are harmless.

High similarity does **not** remove the need for justification.

5. **They are expensive and time-consuming**

   Collecting huge amounts of real-world data and building detailed models often requires major resources.

6. **Too much data can hide important patterns**

   When everything is included, it becomes harder to identify what really matters causally.

Isolation model

What is it?

An isolation model focuses on **only a few important factors** and ignores the rest.

It is simple and studies one part of a complex system in isolation, instead of trying to mirror everything.

Example:

The Wright brothers tested **lift**, **propulsion**, and **control** separately using simple models before building a full airplane.

A basic economic model that studies supply and demand alone, ignoring all other market details, is also an isolation model.
